Buying the Pullback in the Cleanest Trends: An EMA Band Strategy for the Current US Equity Regime

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ABSTRACT

We study a long-only US equity strategy that holds five stocks with the cleanest trends, buys each when its price pulls back into the band between the 20-day exponential moving averages of opens and closes, and sells after 63 trading days. Under the QISC cost model, with fees, square-root market impact and every order capped at 5% of the stock's daily traded value, it earns 22.1% a year from January 2015 to November 2026 against 13.5% for the S&P 500, turning $1 into $10.6 against $4.5, with a Sharpe ratio of 0.75 against 0.81 and a worst drawdown of 43%. The result depends on the market regime. The same rules lose 1.1% a year over 2007–2014, and since 2015 they have done best in trending markets with many clean trends and a few extreme winners. Pre-registered tests show that market trend, the supply of clean trends and small-cap leadership all point the right way in both decades, but none is significant, and switching out of the strategy after a poor year would have destroyed most of its return. In the 38 trading days since the rules were fixed, the portfolio has lost 7.9% against 0.7% for the index, a shortfall seen in one of five comparable windows. One component holds in every sample: entries on a band touch beat entries into the same trends without one by 1.6 percentage points over five days and 5.1 points over three months.

Keywords: Momentum; Short-term reversal; Exponential moving average; Market regimes; Strategy decay; Out-of-sample testing

Submitted to the Quant Investment Strategy Competition (QISC) 2027, Preliminary Round. Every number, table and figure in this paper is produced by `python run.py` in the accompanying code repository (commit `[hash]`). The regime tests in Section 7 follow a protocol committed to the repository before they were run.

1. Introduction

Buying a strong stock on a dip is among the most widely practised trading ideas and among the least rigorously tested. It combines two well-documented regularities. Over horizons of months, returns continue: stocks that have risen tend to keep rising (Jegadeesh and Titman, 1993; Moskowitz et al., 2012). Over days to weeks, returns partly reverse (Jegadeesh, 1990; Lehmann, 1990). An investor who selects strong trends but times the purchase to a short pullback should, in principle, earn the first premium at a better price because of the second.

We turn this idea into a fully specified, point-in-time strategy and evaluate it as the competition requires it to be traded. Each day we rank US stocks by the quality of their current trend, measured as the trend's return divided by its typical pullback. Among the five top-ranked stocks, we buy one when its price dips into the band between the 20-day exponential moving average (EMA) of opens and of closes and then closes back above it, and we sell every position after 63 trading days. Trading costs follow the QISC cost model, with fees and square-root market impact, and every order is capped at 5% of the stock's daily traded value. The main sample is the casebook's backtest period, January 2015 to November 2026. Because the rules were developed on data from July 2017 onward, we add two pieces of evidence that played no role in their design: the same rules over 2007–2014, and a forward test from 19 September 2026, when the rules were fixed.

The strategy works in the current regime. It earns 22.1% a year against 13.5% for the S&P 500, beats the index in seven of twelve calendar years and has gained 196% in 2026 to date. Its Sharpe ratio, 0.75, is close to the index's 0.81 and above what the best of the 51 variants we test would show by chance, but its monthly excess return is not statistically significant, the forward test has so far lost 7.9% against 0.7%, and over 2007–2014 the same rules lose money. We therefore study the regime directly. Since 2015 the market has trended more often, the screen has found more clean trends, and the winners have been larger. In tests specified before we ran them, three real-time indicators of this regime point the right way in both decades, but none is individually significant, and a rule that steps aside after a poor year cuts the 2007–2026 return from 10.9% to 3.8% a year. The timing rule, by contrast, holds in every sample: within the same month, touch entries beat entries into the same trends without one.

Section 2 sets out the economic rationale and hypotheses, Section 3 the data and sample design, and Section 4 the strategy and evaluation methods. Section 5 reports the results, including the forward test, Section 6 the robustness tests, and Section 7 the regime analysis and how we would detect decay. Section 8 covers implementation and risk, and Section 9 concludes.

2. Economic rationale and hypotheses

2.1. *Mechanism*

We classify the edge as behavioural, with a liquidity component. Momentum is commonly attributed to investors who underreact to gradually arriving information (Jegadeesh and Titman, 2001), and the disposition effect, the tendency to sell winners too early, is one channel through which such underreaction arises (Shefrin and Statman, 1985; Frazzini, 2006). Short-term reversal,

in contrast, is largely compensation for supplying liquidity to investors who need to trade (Lehmann, 1990; Nagel, 2012). A pullback within a strong trend is where the two effects meet: the trend is intact, but the price has been pushed temporarily below it by selling unrelated to the firm's prospects. The counterparty on the entry day is a holder who sells into the dip, such as a trader taking profits, an investor selling a winner too early, or a trend-follower whose stop has been triggered. On the exit day, three months later, the typical counterparty is a later momentum buyer.

2.2. *Related evidence*

Technical trading rules have a long and contested record. Moving-average rules appeared profitable on the Dow Jones index (Brock et al., 1992), but much of that evidence weakens once the number of rules searched is accounted for (Sullivan et al., 1999) and once transaction costs and false discoveries are controlled (Bajgrowicz and Scaillet, 2012). At the intraday horizon, none of 7,846 popular rules survives a data-snooping correction in US equities (Marshall et al., 2008). Cross-sectional moving-average signals have held up better (Han et al., 2013), and algorithmically identified retracement zones predict price bounces, although a bounce does not by itself imply a profitable strategy (Tsinaslanidis et al., 2022). Machine-learning studies routinely use moving averages as inputs (Patel et al., 2015; Sezer and Ozbayoglu, 2018) but rarely evaluate a complete, cost-aware portfolio. Momentum profits depend on the state of the market: they are concentrated after rising markets (Cooper et al., 2004) and crash in sharp rebounds after sell-offs (Daniel and Moskowitz, 2016). Factor returns themselves tend to continue (Ehsani and Linnainmaa, 2022), which would justify timing a strategy by its own recent performance. Against all of this, published anomalies decay (McLean and Pontiff, 2016), and rules selected after inspecting a sample overstate their future performance (Harvey et al., 2016).

2.3. *Hypotheses*

The strategy makes a selection bet and a timing bet, and its performance may depend on the regime. We test four hypotheses.

H1 (selection). Stocks ranked in the top five by trend quality earn a positive return in excess of the equal-weighted eligible universe over the following 63 trading days.

H2 (timing). Within the same month, entries into top-ranked stocks on a band touch earn more over the following 5, 21 and 63 trading days than entries into top-ranked stocks without a touch.

H3 (portfolio). A five-position portfolio that combines both rules earns a positive return in excess of the S&P 500 after costs over 2015–2026 and in the QISC out-of-sample window.

H4 (regime). The portfolio's excess return differs between market regimes, and the regime can be identified in real time.

3. Data and sample design

3.1. *Universe and prices*

The universe starts from every security in the NASDAQ Trader symbol directory (November 2026) listed on NASDAQ, NYSE, NYSE American or NYSE Arca, excluding exchange-traded funds, test

issues, warrants, rights, units, preferred shares and notes. Using the country and industry fields of the Nasdaq stock screener, we keep companies located in the United States and drop closed-end funds, which leaves 4,231 names. Daily open, high, low and close prices and volume come from Yahoo Finance, adjusted for splits and dividends, from January 2005 to 11 November 2026. On an average day about 40 stocks pass the trend screen. The benchmark is SPY with dividends reinvested. Zero and negative prices are set to missing, as is any open more than three times above or below the same day's close; no price is forward-filled. We focus on the US because the screen needs a large universe to find enough clean trends on any given day.

3.2. *Sample design*

The rules were developed on data from July 2017 to 18 September 2026. Table 1 shows how the sample is divided and what each period can tell us.

Table 1. Sample periods.

Period	Dates	Role
Main sample	2 Jan 2015 to 11 Nov 2026	The casebook's backtest period; all main results
In-sample	2 Jan 2015 to 31 Dec 2024	Overlaps the development data from July 2017
QISC out-of-sample	2 Jan 2025 to 11 Nov 2026	The final 24 months required by the casebook; overlaps the development data up to September 2026
Forward test	19 Sep 2026 to 11 Nov 2026	After the rules were fixed; the only period no design decision could have seen
Before the sample	3 Jan 2007 to 31 Dec 2014	Never used in development; the same rules in a different regime

We report the QISC window separately, as the casebook requires, but it is not a clean out-of-sample test of these rules. The forward test is clean but short, and the years before the sample are clean but come from a different market. Section 7 uses both to ask whether the strategy's performance is a property of the current regime.

3.3. *Survivorship bias*

The symbol directory lists only stocks that trade today, so delisted companies are missing. The bias favours the strategy, because the missing stocks are disproportionately those whose trends ended badly, and it grows with distance from the present. Results before 2015 are therefore an upper bound, which strengthens any negative conclusion drawn from them.

4. Strategy and evaluation

4.1. *The EMA band*

For a price series x_t, the exponential moving average with span N is

$$E_t = \alpha x_t + (1-\alpha)E_{t-1} = \sum_{k\geq 0} \alpha(1-\alpha)^k x_{t-k}, \qquad \alpha = \frac{2}{N+1}. \tag{1}$$

With $N = 20$, today's price receives 9.5% of the weight and the weight halves every 6.9 trading days, so E_t approximates the average price paid by the last month's buyers. We compute the EMA on opens (E_t^O) and on closes (E_t^C) and call the space between them the band, with upper edge $U_t = \max(E_t^O, E_t^C)$ and lower edge $L_t = \min(E_t^O, E_t^C)$. A close above the band is above where recent buyers both started and ended the day (Figure 1).

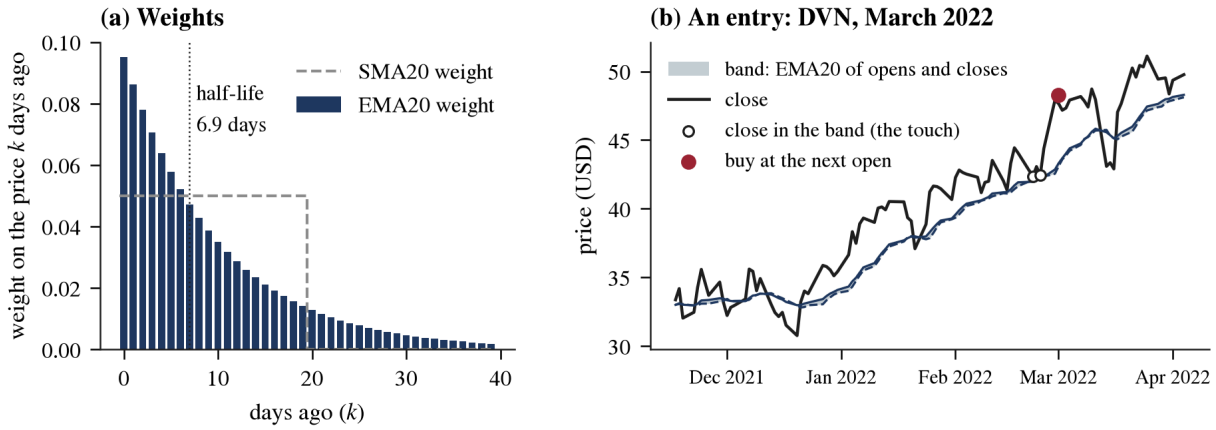


Figure 1. (a) The weight that the EMA20 places on the price k days ago, compared with a 20-day simple moving average. (b) An entry in the portfolio: the close falls into the band, then closes back above it, and the stock is bought at the next open.

4.2. *Trend selection*

A run starts on day $r(t)$, the first day after the latest stretch of more than ten consecutive closes below E^C. A stock passes the screen on day t if its close is above E_t^C and above the 100-day EMA of closes; the run has lasted at least 60 trading days and gained at least 5% in log terms; it contains at least two completed pullbacks of 5% or more, identified on closes with a zigzag filter; no single day contributes more than 20% of the run's log return and no five days more than 55%; and the price and liquidity floors are met. Stocks that pass are ranked by

$$\mathrm{R/R}_t = \frac{C_t/C_{r(t)} - 1}{|\bar{D}_t|}, \tag{2}$$

where C_t is the close and $\bar{D}_t$ the average depth of the completed pullbacks in the current run: the number of its own typical pullbacks the stock has gained. All quantities use prices up to day t, with one exception: a pullback whose low lies before t is counted even if the 5% rebound that confirms it occurs after t. Section 6.4 removes this exception.

4.3. *Entry, exit, sizing and costs*

A top-ranked stock generates an entry signal when, within the last five trading days, it closed inside the band or less than 1% above it, and today closes above the band:

$$\min_{s\in[t-4,\,t]} \left(\frac{C_s}{U_s} - 1\right) \leq 1\% \quad \text{and} \quad C_t > U_t. \tag{3}$$

Decisions are made at the close of day t and executed at the open of day $t+1$. Each new position receives one fifth of current equity, capped by available cash and by 5% of the stock's 60-day median traded value, measured on a notional book that starts at $1 million and compounds. Every position is sold at the open 63 trading days after entry, and a stock sold in the previous five days cannot be bought again. Each order pays fees of 0.05% and market impact of $\max\left(0.05\%,\ 1\% \times \sqrt{Q/\mathrm{ADV}}\right)$, where Q is the order's value and ADV the 60-day median traded value. Uninvested cash earns nothing and the portfolio is not levered. Table 2 lists every parameter.

Table 2. Parameters of the portfolio.

Parameter	Value	Definition
Universe	US companies, no funds	Price at least $1; 60-day median traded value at least $0.5 million
Trend	EMA20 and EMA100	Close above the 20-day and the 100-day EMA of closes
Run	$>$ 10 closes; 60 days; +5%	Closes below EMA20 that end a run; minimum length and log return
Pullbacks	2 of at least 5%	Zigzag reversals on closes within the current run
Gap filter	20% / 55%	Maximum share of the run's log return from the best day / best five days
Ranking	Top 5 by R/R	Eq. (2)
Entry	5 days, 1%	Window and tolerance of the band touch, Eq. (3)
Positions	5 × 20% of equity	Long-only, unlevered
Liquidity	5% of ADV	Maximum order size, on the compounding notional book
Exit	63 trading days	Sold at the open; no other exit
Costs	Fees + impact	0.05% fees; impact max(0.05%, 1% × √(Q / ADV)) per side

4.4. *Evaluation*

We report the compound annual growth rate (CAGR), annualised volatility, Sharpe and Sortino ratios, maximum drawdown, and beta and annualised alpha from a regression of monthly returns on SPY. Sharpe ratios are computed from daily returns without subtracting a risk-free rate, for the strategy and the benchmark alike. Confidence intervals for the monthly excess return resample whole months 2,000 times. In the event study, a touch event is the first day of an entry signal for a top-ranked stock and a no-touch event is a day on which a top-ranked stock trends above the band without a recent touch, which corresponds to buying immediately. Returns run from the next open, net of 0.10% per side, and are also measured in excess of the equal-weighted eligible universe. Because no-touch events cluster in strong markets, we compare the two entries within each month and average the monthly differences.

The regime tests in Section 7 follow a protocol that we committed to the repository before running them. It fixes six indicators, each computed at a month-end from data available then and applied to the following month, the criterion for calling a regime identified, and the switching rules. Every variant in Sections 6 and 7 counts as a trial for the deflated Sharpe ratio of Bailey and López de Prado (2014).

5. Results

5.1. *The portfolio, 2015–2026*

Table 3 and Figure 2 report the portfolio over the main sample and by period. Over 2015–2026 it earns 22.1% a year against 13.5% for SPY and ends at 10.6 times its starting value against 4.5 times. Its Sharpe ratio, 0.75, is slightly below the index's 0.81, and its monthly excess return of 0.97% has an interval that includes zero. In the in-sample years it earns 15.7% a year against 13.0%; in the QISC out-of-sample window it earns 28.3% against 16.1%, with a drawdown of 42% against 19%. Before the sample it loses money, with a drawdown of 64% from October 2007 to October 2008. The excess return is positive in both periods of the main sample but distinguishable from zero in neither, so H3 is supported in direction but not in significance. Table A.1 adds two comparisons: over the development window alone the portfolio earns 28.4% a year, and a diversified ten-position version earns 16.7% over 2015–2026.

Table 3. Performance of the portfolio: main sample, in-sample years, QISC out-of-sample window and the years before the sample.

Measure	2015–2026	2015–2024	2025–2026	2007–2014
Panel A: Return				
CAGR	+22.1%	+15.7%	+28.3%	−1.1%
SPY CAGR	+13.5%	+13.0%	+16.1%	+7.0%
Monthly excess return	+0.97%	+0.44%	+1.68%	−0.50%
95% interval	[−0.50%, +2.54%]	[−0.86%, +1.85%]	[−3.68%, +7.36%]	[−1.79%, +0.77%]
Alpha per year	+10.7%	+4.4%	+23.7%	−5.5%
Cumulative return (SPY)	+963% (+347%)	+330% (+240%)	+59% (+32%)	−8% (+72%)
Panel B: Risk				
Volatility per year	34.2%	30.5%	54.5%	28.7%
Sharpe ratio (SPY)	0.75 (0.81)	0.63 (0.78)	0.73 (0.96)	0.11 (0.42)
Sortino ratio	1.04	0.86	1.12	0.13
Worst drawdown (SPY)	−43% (−34%)	−43% (−34%)	−42% (−19%)	−64% (−55%)
Beta	1.10	1.07	0.91	0.97
Information ratio	0.37	0.20	0.43	−0.27
Panel C: Trades				
Trades	220	186	31	138
Win rate	52%	51%	55%	50%
Average win / loss	+35.3% / −16.8%	+27.9% / −16.1%	+52.4% / −23.3%	+18.6% / −19.2%
Mean / median trade	+10.4% / +2.5%	+6.4% / +1.4%	+18.2% / +3.0%	−0.3% / −0.3%

Note: Each column is a separate portfolio that starts in cash. Intervals resample whole months. Alpha and beta are from a regression of monthly returns on SPY. Annual returns are in Table A.3.

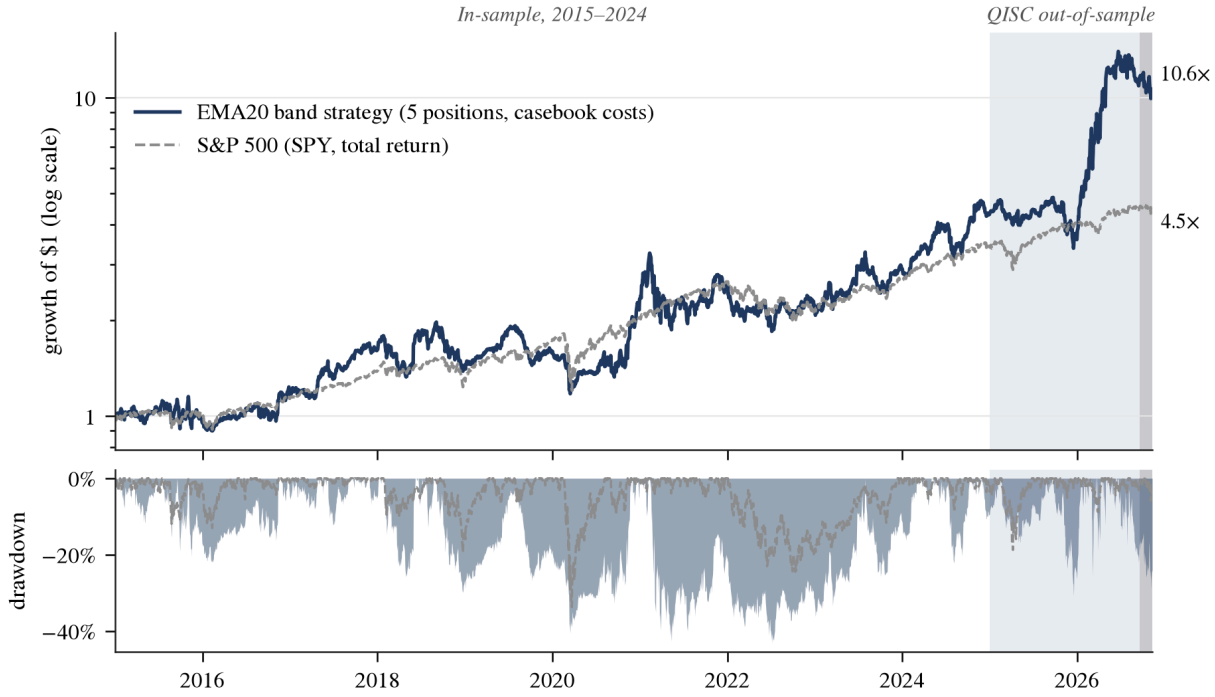


Figure 2. Growth of $1 on a log scale (top) and drawdown from the previous peak (bottom), portfolio and SPY, January 2015 to November 2026. Light shading marks the QISC out-of-sample window and dark shading the forward test.

The portfolio moved broadly with the index until 2023, sometimes ahead and sometimes behind, and pulled clearly ahead in 2024 and 2026 (Figure 2). It beats SPY in seven of twelve calendar years, and its returns are concentrated in a few years and a few names: 2020 (+46.0%), 2024 (+47.2%) and 2026 to date (+196.4%). Trade returns are strongly skewed (Figure A.3): the median trade earns 2.5% and the mean 10.4%, eleven of 220 trades more than doubled, and 19 lost at least 30%. The ten best trades account for 74% of summed trade returns.

Realistic frictions matter for a concentrated strategy (Table A.2). With a flat 10 basis points per side and no limit on order size, the same rules earn 30.4% a year and their excess return is significant. Square-root impact costs remove 1.4 points, and capping each order at 5% of daily traded value removes about 7 more, because the strongest trends are often in stocks too thin to absorb a fifth of a growing portfolio.

5.2. *2026 and the forward test*

The portfolio has gained 196.4% in 2026 to date against 11.7% for SPY (Figure 3a). Most of the gain came between January and June, from a handful of semiconductor, storage and optical-networking stocks, and the portfolio has since fallen as much as 29% from its peak.

The rules were fixed on 18 September 2026. In the 38 trading days since, the portfolio has lost 7.9% while SPY lost 0.7% (Figure 3b). Six positions were held over the period (Table C.3): three lost about 24% each, three gained between 3% and 7%, and only one was opened after the rules were fixed. A shortfall of 7.2 points over 38 days is not unusual for this strategy: one in five 38-day windows in 2015–2026 was at least as bad. The forward test is therefore negative but not yet informative; the live evaluation that the competition runs after the submission deadline will extend it.

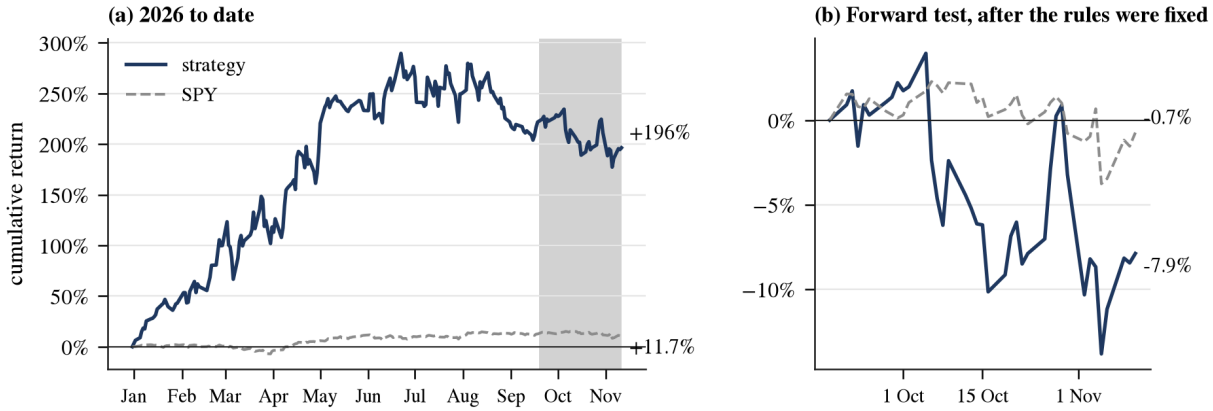


Figure 3. (a) Cumulative return of the portfolio and SPY in 2026 to date; the shaded region is the forward test. (b) The forward test: cumulative return since the close of 18 September 2026, when the rules were fixed.

5.3. *The timing effect*

Table 4 reports the event study. Over 63 trading days, touch entries earn 4.7 percentage points more than the equal-weighted universe and no-touch entries into the same top-ranked trends 4.1 points more; both intervals exclude zero, so H1 is supported in 2015–2026. Before the sample, neither group beats the universe.

Table 4. Event study: entries into top-five stocks with and without a band touch.

Sample and horizon	Touch: above universe [95%]	No touch: above universe [95%]	Touch minus no touch, same month [95%]
Panel A: 2015–2026			
5 days	+0.3% [−0.3%, +0.9%]	−0.4% [−0.8%, +0.1%]	+1.6% [+0.7%, +2.6%]
21 days	+1.8% [+0.4%, +3.4%]	+0.3% [−1.1%, +1.7%]	+3.2% [+1.7%, +4.9%]
63 days	+4.7% [+1.7%, +7.9%]	+4.1% [+1.0%, +7.5%]	+5.1% [+1.7%, +8.9%]
Panel B: By period, 5 days			
2015–2024	+0.3% [−0.3%, +1.0%]	−0.4% [−0.8%, −0.0%]	+1.8% [+0.9%, +2.9%]
2025–2026	+0.1% [−1.5%, +1.9%]	−0.0% [−1.4%, +1.2%]	+0.4% [−1.8%, +2.4%]
2007–2014	−0.5% [−1.2%, −0.0%]	−0.2% [−0.6%, +0.3%]	+0.7% [+0.0%, +1.3%]
Panel C: By period, 63 days			
2015–2024	+4.1% [+1.6%, +6.8%]	+2.0% [−0.5%, +4.2%]	+4.7% [+1.6%, +7.9%]
2025–2026	+8.6% [−5.5%, +26.8%]	+15.8% [−0.4%, +32.6%]	+7.7% [−5.8%, +26.5%]
2007–2014	−0.7% [−3.5%, +2.2%]	−1.4% [−3.7%, +1.0%]	+2.6% [−0.8%, +5.7%]

Note: 1,327 touch and 9,570 no-touch events in 2015–2026. Returns run from the next open, net of 0.10% per side. "Above universe" subtracts the equal-weighted return of all eligible stocks over the same days. The last column is the mean over months of the within-month difference between touch and no-touch returns; it compares raw returns.

The timing effect (H2) is the most robust result in the paper. Within the same month, touch entries beat no-touch entries by 1.6 percentage points over five days, 3.2 over 21 days and 5.1 over 63 days, with every interval excluding zero. Before the sample the effect is smaller but still positive at five and 21 days (+0.7 and +1.5 points, both intervals just above zero), and it has the same sign

in the out-of-sample window, where less than two years of events leave the intervals wide. In pooled averages (Figure A.4) the gap is smaller, 0.7 points at 63 days, because no-touch entries cluster in strong markets.

6. Robustness

6.1. *Exit rules*

Table B.1 compares eight alternative exits on the same entries. None improves on the 63-day clock. Exits on closes below the band earn 4.0% to 15.4% a year; stop-losses at 25% and 40% earn about 17%, consistent with the mixed evidence on stop-losses in long-only portfolios (Lei and Li, 2009; Kaminski and Lo, 2014); and an exit on a heavy-volume down day earns 22.8% with a deeper drawdown of 53%.

6.2. *Position count, holding period and costs*

Table B.2 and Figure B.1 vary one parameter at a time. Concentration pays in this sample: three to five positions earn 22% to 25% a year and eight or more 12% to 17%, with no smaller drawdown; before 2015, more positions reduce losses. Holding periods of 42 to 84 days earn 21% to 22%, and 21 or 126 days about 16%. The portfolio trades about 7.4 times its capital a year, counting purchases and sales, at an average cost of 0.18% per side, so doubling the cost model lowers the CAGR to 21.3% and five times the cost model to 18.9%.

6.3. *Entry parameters*

Table B.3 varies the touch window, the tolerance above the band and the EMA span. Requiring a close strictly inside the band raises the CAGR to 27.0%, and the strict touch is the best tolerance for every window. The pattern is not monotonic, however: a 1% tolerance is worse than both 0% and 2% in every window. A five-position portfolio is path-dependent, since one missed entry changes every later one, so we read the grid as noise around a positive effect and do not switch to the strict touch after seeing it. Spans from 10 to 40 days give CAGRs between 15.1% and 22.1%, with the 20-day span at the top, as expected of a parameter chosen on these data.

6.4. *Stress tests*

Table B.4 reports the stress tests described in the casebook and three further checks. Delaying every trade by one day lowers the return to 20.2% a year, and counting pullbacks only once confirmed to 21.1%. Removing the five largest contributors (AXTI, SNDK, NUTX, APPS and CELH) lowers it to 12.0%, below the index. Buying the same stocks without waiting for a touch earns 17.3% with a drawdown of 57%, and loses 12.3% a year in the out-of-sample window against a gain of 28.3% for the portfolio: the timing rule is what separates the two. Starting the portfolio in different months of 2015 gives CAGRs between 18.3% and 26.7%.

6.5. *Multiple testing*

Sections 6 and 7 report 51 variants, counting the baseline each time it appears. If none had any true edge, the expected maximum annualised Sharpe ratio across 51 trials with the observed

dispersion would be 0.24. The deflated Sharpe ratio of the portfolio, the probability that its true Sharpe ratio exceeds that benchmark after allowing for skewness and fat tails, is 0.961, above the conventional 0.95. The portfolio's risk-adjusted return is therefore unlikely to be an artefact of the variants we tried; it is not, however, higher than the index's.

7. Regime and decay

7.1. *What changed after 2015*

Figure 4 runs the same rules continuously from 2007 and plots the monitors defined in the protocol. The portfolio's three-year return relative to SPY was negative at nearly every month-end from 2010 to 2018, positive for short spells in 2019 and 2021, and positive almost continuously since late 2022. Table 5 shows how the market differed between the periods.

Table 5. Market conditions and the portfolio's trades by period.

Measure	2007–2014	2015–2024	2025–2026
Stocks passing the trend screen per day	35	44	39
Days with SPY above its 200-day average	64%	82%	88%
Equal-weighted universe minus SPY, per year	+3.7%	−3.6%	−8.5%
Trades	143	189	29
Trades in technology / health care	6% / 15%	16% / 17%	21% / 24%
Median trade	+0.7%	+1.4%	+6.7%
Trades returning +50% or more	3.5%	9.0%	17.2%
Touch minus no touch at 21 days, same month	+1.5%	+3.0%	+4.4%

Note: Trades are from the same rules run continuously from 2007. Sectors are from the Nasdaq stock screener.

Three differences stand out. The market trended more: SPY spent 82% to 88% of days above its 200-day average after 2015, against 64% before, a period that includes the 2008 crisis and the 2011 sell-off. The screen found more clean trends, and the trades it produced were more often extreme winners: 9% to 17% of trades returned 50% or more, against 3.5% before. And the timing effect itself was two to three times stronger. The current regime is not one of broad small-cap strength, since the equal-weighted universe has lagged SPY since 2015; it is a trending market in which a few stocks, increasingly in technology and health care, make very large moves.

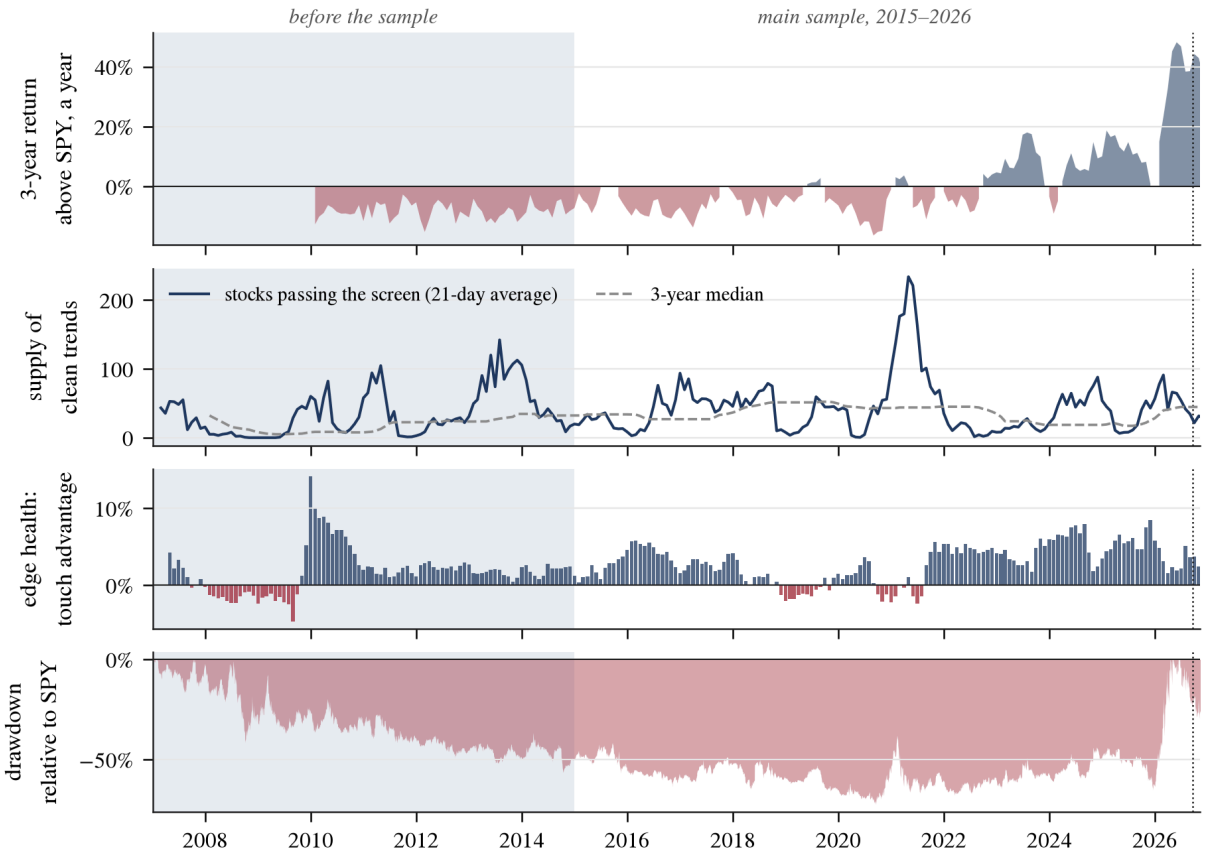


Figure 4. The regime and decay monitors, 2007 to November 2026, for the same rules run continuously from 2007: (a) three-year return above SPY, per year; (b) the number of stocks passing the trend screen and its three-year median; (c) edge health, the month-matched touch advantage at 21 days over the past year; (d) the portfolio's drawdown relative to SPY. Shading marks the years before the sample and the dotted line the start of the forward test.

7.2. *Can the regime be identified in real time?*

Table 6 reports Test 1 of the protocol: the portfolio's return above SPY in the month after each indicator is on, minus the same return after it is off. The protocol calls a regime identified if the difference is positive both before and within the sample and the pooled interval excludes zero.

Table 6. Next-month return above SPY with each regime indicator on, minus the same with it off.

Indicator	2007–2014 [95%]	2015–2026 [95%]	Pooled [95%]
R1 Market trend	+1.72% [−1.64%, +5.23%]	+0.49% [−2.48%, +3.41%]	+1.26% [−1.01%, +3.57%]
R2 Supply of clean trends	+1.36% [−1.31%, +4.08%]	+1.88% [−1.17%, +5.00%]	+1.63% [−0.59%, +3.78%]
R3 Small-cap leadership	+1.14% [−1.59%, +3.95%]	+2.28% [−1.15%, +6.16%]	+1.30% [−0.99%, +3.52%]
R4 Strategy momentum	−5.78% [−8.28%, −3.34%]	−1.03% [−4.14%, +2.14%]	−2.05% [−4.24%, +0.02%]
R5 Edge health	+1.62% [−2.28%, +5.15%]	−0.79% [−6.26%, +3.99%]	+0.52% [−2.94%, +3.61%]
R6 R4 and R5 both on	−5.08% [−7.53%, −2.45%]	−0.44% [−3.71%, +2.74%]	−1.22% [−3.61%, +1.04%]

Note: Percentage points per month; intervals resample months within each group. On when: R1, SPY is above its 200-day average; R2, the number of stocks passing the screen is above its 3-year median; R3, the equal-weighted universe has beaten SPY over 6 months; R4, the portfolio has beaten SPY over 12 months; R5, the month-matched touch advantage was positive over the past year. No indicator meets the protocol's criterion.

No indicator meets the criterion, so H4 is not supported in its strict form. Three indicators point the right way in both decades: the portfolio did 0.5 to 2.3 points a month better after the market trended (R1), after the supply of clean trends was above its median (R2) and after small stocks led (R3). With about 85 months before the sample and 143 within it, none of these differences is significant on its own. The strategy's own recent performance points the wrong way: after a year in which it beat SPY, it did worse in the following month, by 5.8 points before the sample, where the interval excludes zero, and by 1.0 point within it. Factor returns in general continue (Ehsani and Linnainmaa, 2022), but this strategy's returns reverse, plausibly because a good year for a five-stock book is made by a few large winners that are then sold.

7.3. *Walk-forward switching rules*

Table 7 reports Test 2: each rule holds the portfolio in a month when its indicator was on at the previous month-end and SPY otherwise. Every decision uses data available at the time, no threshold is fitted, and each switch costs 0.25%.

Table 7. Walk-forward rules that hold SPY instead of the portfolio when an indicator is off.

Rule	CAGR, 2007–2014	CAGR, 2015–2026	CAGR, 2007–2026	Sharpe, 2007–2026	Worst drawdown	Monthly excess [95%]
Always invested	−1.1%	+19.9%	+10.9%	0.48	−64%	+0.33% [−0.73%, +1.38%]
R1 Market trend	+4.8%	+18.5%	+12.8%	0.53	−56%	+0.43% [−0.53%, +1.42%]
R2 Clean trends	+4.5%	+20.9%	+14.0%	0.57	−63%	+0.50% [−0.39%, +1.44%]
R3 Small caps	+4.0%	+21.4%	+14.1%	0.62	−63%	+0.39% [−0.31%, +1.17%]
R4 Momentum	−8.6%	+13.1%	+3.8%	0.27	−74%	−0.43% [−1.12%, +0.23%]
R5 Edge health	+4.0%	+17.7%	+12.0%	0.52	−53%	+0.33% [−0.54%, +1.27%]
R6 R4 and R5	−4.6%	+14.4%	+6.3%	0.36	−64%	−0.23% [−0.89%, +0.39%]
SPY	+7.0%	+13.5%	+10.8%	0.62	−55%	

Note: The portfolio runs continuously from 2007 as a shadow book, so its returns in 2015–2026 differ slightly from Table 3, where it starts in cash in January 2015.

The three structural indicators help. Each turns the years before the sample from a loss of 1.1% a year into a gain of 4.0% to 4.8%, still below SPY's 7.0%, and leaves 2015–2026 roughly unchanged; over 2007–2026 they raise the CAGR from 10.9% to between 12.8% and 14.1%. The small-cap rule brings the Sharpe ratio up to the index's, 0.62. None makes the excess return significant. The rules based on the strategy's own performance destroy value: stepping aside after a year behind SPY cuts the 2007–2026 return to 3.8%.

7.4. *Detecting decay*

The tests suggest how to decide when the strategy has stopped working. First, do not switch off on recent performance. Rules R4 and R6 would have taken the strategy out after its losing years and put it back after its winning years, the wrong way round in both decades. A drawdown measured against the index is no better: the continuous book fell 41% behind SPY by 2009 and 72% behind by 2020, and regained its 2007 relative peak only in April 2026 (Figure 4d), so a relative-drawdown stop would have ended the strategy before every one of its good years. Second, monitor the conditions the strategy needs. Market trend (R1) and the supply of clean trends (R2) measure whether the market is producing the trends the screen looks for; when both were off, as in 2008, late 2011, late 2015 to early 2016 and most of 2022, the strategy had little to work with. Third, monitor the mechanism. Edge health (R5) measures whether the timing effect still works. It was negative in 2008–2009 and briefly in 2019–2021, and has been positive since 2022. It reacts too slowly to time the portfolio, but a year of negative readings would mean that the pullback entry has stopped adding value, the clearest sign of decay this strategy can give.

Because no indicator met the protocol's criterion, we do not add a filter to the submitted strategy. We would report the three monitors monthly and stop trading if edge health is negative for twelve consecutive months or the drawdown exceeds 55%, deeper than any start month in 2015 produced (47% at worst).

7.5. *How the regime could end*

Three developments would end the current regime. A prolonged bear or range-bound market, as in 2008 or 2011–2012, would reduce the number of clean trends and break more of those that remain; R1 and R2 would show it within a month or two. A reversal of the themes behind the largest winners would hit a five-stock book hard: the 2026 gains came from a few semiconductor, storage and optical-networking stocks, and the portfolio has already fallen as much as 29% from its 2026 peak. And a sharp rebound after a sell-off, when momentum strategies crash (Daniel and Moskowitz, 2016), would hurt a strategy that buys the strongest trends. If pullback buying in strong trends became crowded, the timing effect could also shrink, as published anomalies tend to (McLean and Pontiff, 2016); edge health would show this, slowly.

8. Implementation and risk

Capacity. The median order is 1.8% of the stock's 60-day median traded value, and 28% of orders are cut back by the 5% cap. The median stock bought trades $25.8 million a day and the 5th percentile $1.0 million, where the cap limits a position to about $50,000. The cap binds more as the book compounds and already costs about 7 points a year (Table A.2).

Execution. All trades are market-on-open orders placed after the previous close, at an average cost of 0.18% per side, most of it market impact in the smaller names. Returns therefore combine overnight and intraday components, which respond differently to news (Bahcivan, 2025).

Risk. Beta is 1.10 and volatility 34%, about twice that of SPY. Investors should expect drawdowns of 40% to 65% and about one trade in twelve to lose 30% or more, and without its five largest contributors the portfolio would have trailed the index. Volatility scaling, which helps conventional momentum portfolios (Barroso and Santa-Clara, 2015), is a natural next test.

9. Conclusion

This strategy works in the current regime. Under the QISC rules it earns 22.1% a year from 2015 to 2026 against 13.5% for the S&P 500, with a Sharpe ratio close to the index's and a risk-adjusted return that survives a correction for the 51 variants we tested. It does not work before 2015, and the regime that suits it, a trending market that produces many clean trends and a few extreme winners, is clear in hindsight. In tests we specified in advance, three real-time indicators of that regime point the right way in both decades but none is significant on its own, so we do not claim that the regime can yet be timed. The forward test since the rules were fixed is negative, though not unusually so.

Two results are robust enough to act on. The pullback entry improves on buying the same trends without a pullback in every sample, by about 1.6 points over a week and 5 points over three months since 2015. And the strategy's own recent performance is the wrong signal for deciding when to stop: monitoring the market conditions it needs and the timing effect it relies on is better. We would run the strategy as a concentrated satellite position with those monitors, and treat the competition's live evaluation as its next test.

Statement of Originality and AI Disclosure

We declare that this paper and the accompanying code are the original work of Team Tester, that all sources are cited, and that the submission has not been submitted to any other competition. Generative AI was used to assist with programming the data pipeline and with formatting tables and figures. The strategy, the analysis and the written paper are the team's own work, and the team takes full responsibility for the content.

Code and data availability

The repository `qisc-tester` reproduces every table and figure with one command, `python run.py`. It downloads the symbol directory, the Nasdaq screener and the prices, computes the point-in-time scores, runs every test, and writes the tables, the figures, the trade list and `output/positions.csv` in the format required by the competition. `REGIME_PROTOCOL.md` contains the regime tests as committed before they were run. All data are publicly available; a full run takes about 30 minutes, most of it the price download.

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Appendix A. Additional results

Table A.1 compares the portfolio with the development window, a diversified version and the years before the sample. Table A.2 shows the effect of the casebook's cost model and liquidity cap. Table A.3 lists calendar-year returns of the portfolio from 2015 and of the same rules run continuously from 2007. Figures A.1 and A.2 show cumulative returns against SPY, Figure A.3 the distribution of trade returns, and Figure A.4 the average path after touch and no-touch entries.

Table A.1. The portfolio, the development window, a diversified version and the years before the sample.

Measure	Portfolio, 2015–2026	Development window, 2017–2026	Ten positions, 2015–2026	Same rules, 2007–2014
Trades	220	171	435	138
Win rate	52%	56%	53%	50%
Growth per year (SPY)	+22.1% (+13.5%)	+28.4% (+15.0%)	+16.7% (+13.5%)	−1.1% (+7.0%)
Sharpe ratio (SPY)	0.75 (0.81)	0.85 (0.85)	0.67 (0.81)	0.11 (0.42)
Worst drawdown (SPY)	−43% (−34%)	−45% (−34%)	−43% (−34%)	−64% (−55%)
Monthly return above SPY	+0.97%	+1.38%	+0.40%	−0.50%
95% interval	[−0.50%, +2.54%]	[−0.43%, +3.38%]	[−0.65%, +1.43%]	[−1.79%, +0.77%]
Cumulative return (SPY)	+963% (+347%)	+896% (+262%)	+524% (+347%)	−8% (+72%)

Note: The development window runs from 3 July 2017 to 18 September 2026. Each column is a separate portfolio that starts in cash.

Table A.2. The effect of trading costs and the liquidity cap, January 2015 to November 2026.

Cost and liquidity assumptions	Per year	Worst drawdown	Sharpe	Monthly excess [95%]
Flat 10 bp per side, no liquidity cap	+30.4%	−47%	0.87	+1.72% [+0.04%, +3.50%]
Impact costs, no liquidity cap	+29.0%	−48%	0.84	+1.63% [−0.05%, +3.41%]
Flat 10 bp per side, orders capped at 5% of ADV	+22.5%	−43%	0.77	+1.00% [−0.47%, +2.56%]
Impact costs and 5% of ADV cap (the portfolio)	+22.1%	−43%	0.75	+0.97% [−0.50%, +2.54%]

Table A.3. Calendar-year returns.

Year	Portfolio, from 2015	Same rules, from 2007	SPY	Portfolio minus SPY
2007		+8.8%	+5.3%	
2008		−54.7%	−36.8%	
2009		+32.7%	+26.4%	
2010		+4.7%	+15.1%	
2011		−7.5%	+1.9%	
2012		+4.7%	+16.0%	
2013		+22.4%	+32.3%	
2014		+13.0%	+13.5%	
2015	−4.5%	−11.0%	+1.2%	−5.7%
2016	+23.4%	+5.0%	+12.0%	+11.4%
2017	+36.3%	+39.9%	+21.7%	+14.6%
2018	−9.8%	−18.0%	−4.6%	−5.3%
2019	+10.5%	+12.3%	+31.2%	−20.7%
2020	+46.0%	+57.0%	+18.3%	+27.7%
2021	+10.5%	+9.4%	+28.7%	−18.3%
2022	−13.4%	−17.4%	−18.2%	+4.8%
2023	+30.7%	+34.3%	+26.2%	+4.5%
2024	+47.2%	+51.5%	+24.9%	+22.3%
2025	−16.6%	−16.4%	+17.7%	−34.4%
2026*	+196.4%	+195.4%	+11.7%	+184.7%

Note: The portfolio starts in cash on 2 January 2015 and beats SPY in seven of twelve years; the same rules run continuously from 2007 hold different positions in 2015 and drift apart from it. * 2026 to 11 November.

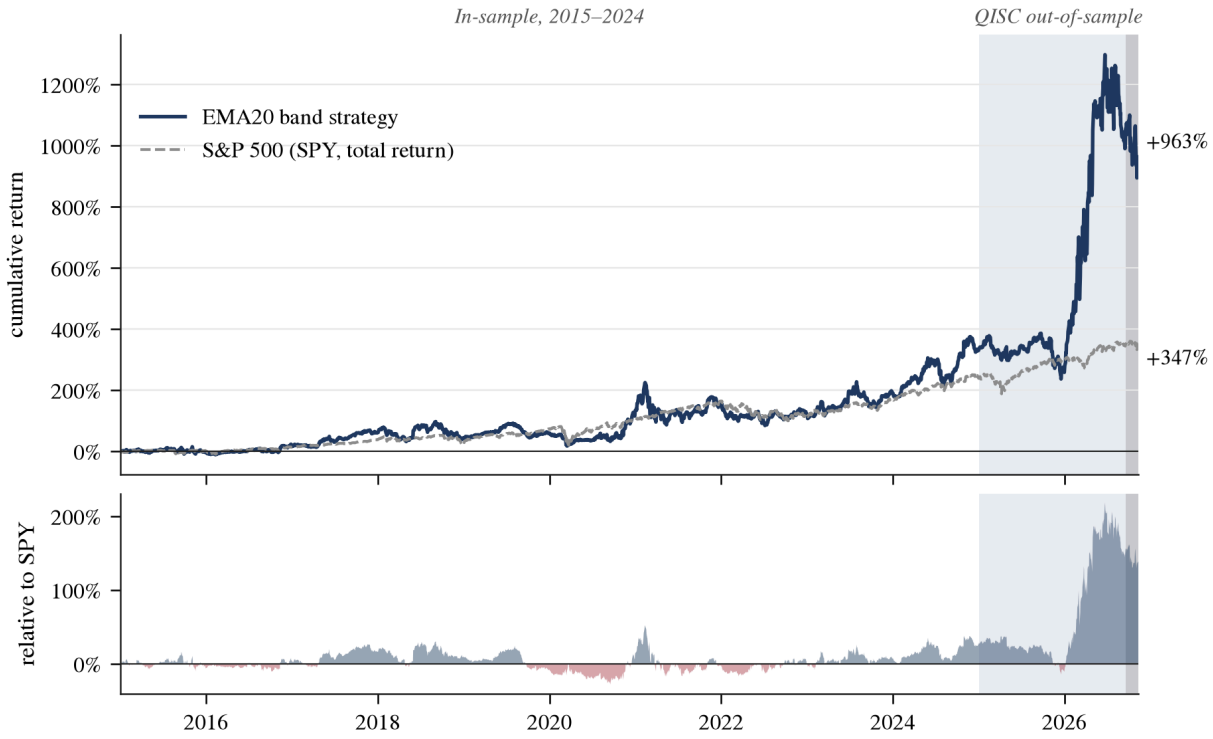


Figure A.1. Cumulative return of the portfolio and SPY, January 2015 to November 2026 (top), and the portfolio's value relative to SPY (bottom). Light shading marks the QISC out-of-sample window and dark shading the forward test.

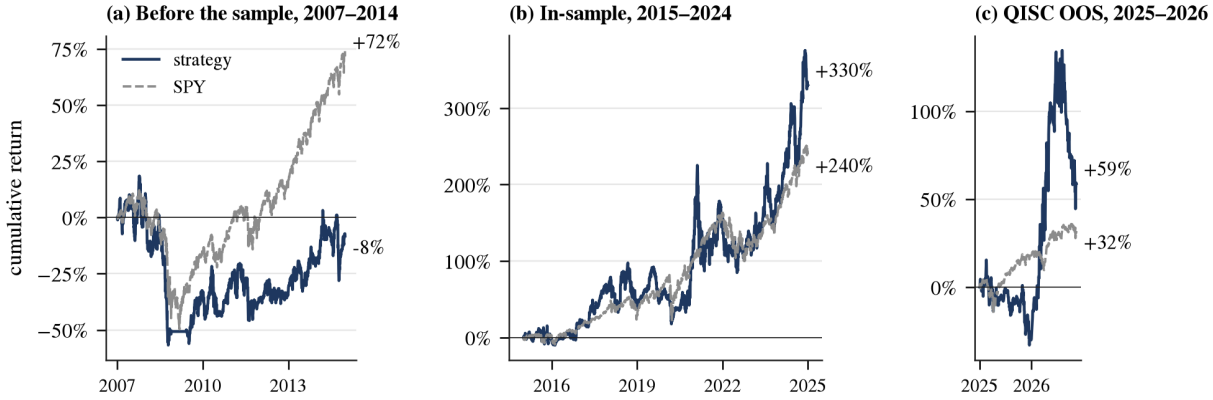


Figure A.2. Cumulative return of the same rules and SPY in each period: (a) before the sample, (b) in-sample, (c) QISC out-of-sample window. Each portfolio starts in cash on the first day of its period, as in Table 3.

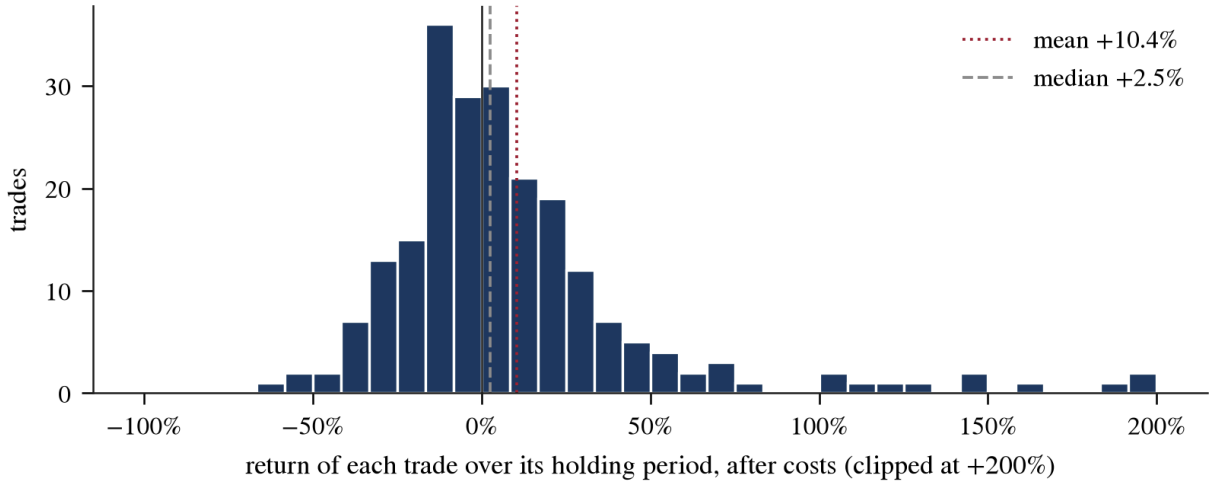


Figure A.3. Return of each trade over its holding period, after costs, 2015–2026. Returns above +200% are shown at +200%.

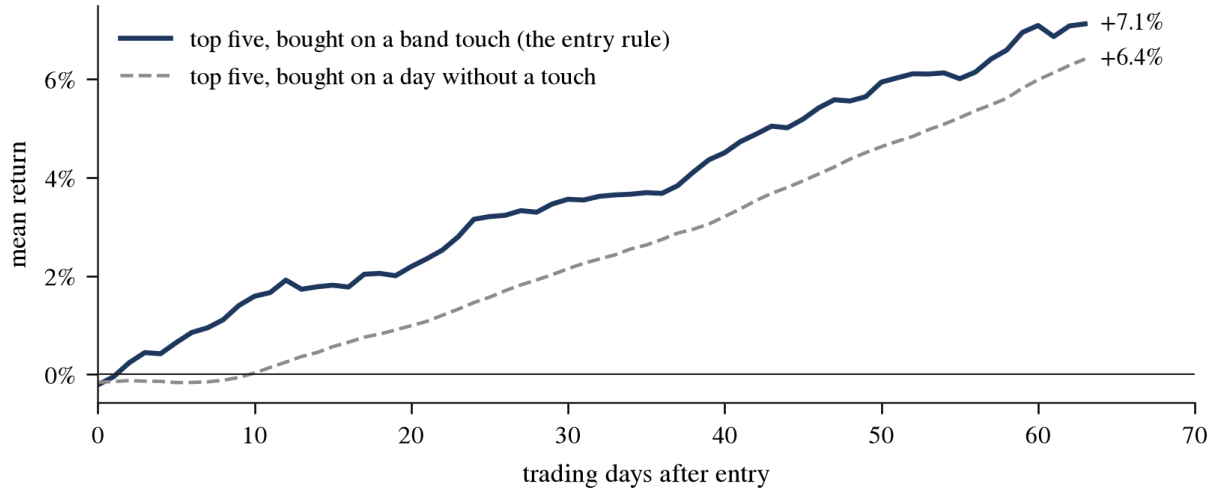


Figure A.4. Mean cumulative return from the entry open for top-five stocks bought on a band touch and without one, 2015–2026, pooled over all events.

Appendix B. Robustness tables

Table B.1. Exit rules, January 2015 to November 2026.

Exit rule (entries unchanged)	Trades	Win rate	Per year	Worst drawdown	CAGR ÷ drawdown	Sharpe
Hold 63 days (baseline)	220	52%	+22.1%	−43%	0.51	0.75
Close below the band	929	31%	+4.0%	−48%	0.08	0.28
Three closes below the band	607	38%	+13.7%	−51%	0.27	0.57
Five closes below the band	483	36%	+15.4%	−47%	0.32	0.62
Heavy-volume down day (≥ 3×)	316	53%	+22.8%	−53%	0.43	0.79
Heavy-volume down day (≥ 5×)	251	53%	+14.6%	−55%	0.27	0.58
Three closes below the band or volume ≥ 3×	673	40%	+15.4%	−50%	0.31	0.63
Stop at −25%	253	49%	+17.1%	−55%	0.31	0.64
Stop at −40%	229	50%	+17.6%	−47%	0.38	0.66

Note: Exits other than the clock are decided on a close and executed at the next open; every position is sold after 63 days at the latest. Heavy-volume down day: a lower close on volume of at least the stated multiple of its 50-day median.

Table B.2. Position count, holding period and costs.

Variant	Trades	Per year	Worst drawdown	Sharpe	CAGR ÷ drawdown: 2015–2026	2015–2024	2025–2026	2007–2014
Panel A: Number of positions								
3	130	+25.4%	−50%	0.81	0.51	0.40	0.72	−0.19
5 (baseline)	220	+22.1%	−43%	0.75	0.51	0.36	0.68	−0.02
8	352	+14.5%	−47%	0.58	0.31	0.22	0.76	0.09
10	435	+16.7%	−43%	0.67	0.39	0.34	0.17	0.13
12	518	+15.3%	−43%	0.64	0.36	0.34	0.39	0.07
15	647	+15.7%	−44%	0.66	0.36	0.29	0.45	0.02
20	846	+11.9%	−49%	0.55	0.24	0.18	0.95	0.17
Panel B: Holding period (trading days)								
21	550	+16.2%	−46%	0.67	0.35	0.33	0.37	0.03
42	317	+20.9%	−52%	0.77	0.40	0.32	0.39	0.14
63 (baseline)	220	+22.1%	−43%	0.75	0.51	0.36	0.68	−0.02
84	169	+22.0%	−46%	0.75	0.48	0.44	0.93	0.00
126	115	+15.8%	−55%	0.61	0.29	0.29	0.10	0.05
Panel C: Cost model multiple								
0.5×	220	+22.4%	−43%	0.76	0.53	0.38	0.69	−0.01
1× (baseline)	220	+22.1%	−43%	0.75	0.51	0.36	0.68	−0.02
2×	220	+21.3%	−44%	0.73	0.48	0.34	0.65	−0.03
3×	220	+20.6%	−46%	0.71	0.45	0.31	0.63	−0.04
5×	220	+18.9%	−48%	0.67	0.39	0.27	0.58	−0.06

Note: Each row changes one parameter of the baseline. Panel C scales both fees and market impact. The last four columns are separate portfolios per period.

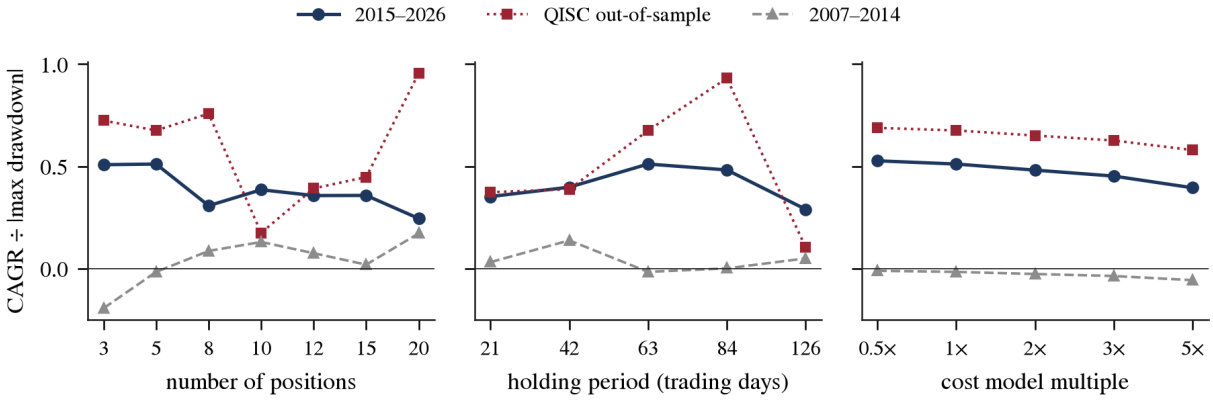


Figure B.1. CAGR divided by the absolute maximum drawdown as one parameter of the portfolio varies, for 2015–2026, the QISC out-of-sample window and 2007–2014.

Table B.3. Entry parameters, January 2015 to November 2026.

Touch window or EMA span	Tolerance	Trades	Per year	Worst drawdown	CAGR ÷ drawdown	Sharpe
Panel A: Touch window						
3 days	0%	203	+16.5%	−43%	0.39	0.61
3 days	1%	215	+12.8%	−45%	0.28	0.53
3 days	2%	221	+14.5%	−44%	0.33	0.57
5 days	0%	212	+27.0%	−38%	0.71	0.90
5 days (baseline)	1%	220	+22.1%	−43%	0.51	0.75
5 days	2%	226	+26.4%	−39%	0.68	0.87
10 days	0%	220	+18.6%	−40%	0.47	0.68
10 days	1%	225	+14.5%	−53%	0.27	0.56
10 days	2%	228	+16.3%	−44%	0.37	0.61
Panel B: EMA span of the band						
10 days	1%	230	+15.1%	−55%	0.27	0.57
15 days	1%	225	+16.4%	−44%	0.38	0.61
20 days (baseline)	1%	220	+22.1%	−43%	0.51	0.75
30 days	1%	198	+20.4%	−46%	0.44	0.73
40 days	1%	176	+16.7%	−46%	0.36	0.65

Note: Panel B keeps the five-day window. The trend screen is unchanged throughout.

Table B.4. Stress tests, January 2015 to November 2026.

Test	Per year	Worst drawdown	Sharpe	Monthly excess [95%]
Portfolio	+22.1%	−43%	0.75	+0.97% [−0.5%, +2.5%]
Execution one day later	+20.2%	−46%	0.70	+0.88% [−0.6%, +2.6%]
Without the five largest contributors	+12.0%	−45%	0.52	+0.10% [−1.1%, +1.4%]
Bought without a touch	+17.3%	−57%	0.61	+0.62% [−0.8%, +2.1%]
Pullbacks counted only once confirmed	+21.1%	−50%	0.73	+0.92% [−0.6%, +2.5%]
New positions only while SPY is above its 200-day average	+14.7%	−52%	0.57	+0.40% [−1.1%, +2.0%]
Twelve start months in 2015	+18.3% to +26.7%	−47% to −42%		
Twelve start months in 2007, to 2014	−4.8% to +5.2%	−69% to −49%		

Appendix C. Trades

The full list of 220 trades is in `output/trades.csv`. Prices are split- and dividend-adjusted opens; returns are after fees and market impact. Open positions are marked at the close of 11 November 2026, before exit costs.

Table C.1. The ten best trades, 2015–2026.

#	Stock	Bought	Buy price	Sold	Sell price	Return
1	AXTI	2026-02-03	21.8	2026-05-05	106.78	+388.2%
2	SNDK	2025-12-11	230.0	2026-03-16	687.52	+198.3%
3	NUTX	2025-02-26	57.0	2025-05-28	165.44	+188.9%
4	CELH	2020-10-28	7.06	2021-01-29	18.66	+163.4%
5	CRDF	2020-09-21	8.9	2020-12-18	21.96	+145.5%
6	LITE	2026-02-02	405.49	2026-05-04	986.88	+142.9%
7	APPS	2020-11-05	39.06	2021-02-08	90.2	+130.4%
8	WDC	2026-03-16	282.63	2026-06-15	618.58	+118.4%
9	WGS	2024-09-11	33.92	2024-12-10	72.41	+112.4%
10	KRMD	2020-01-27	5.89	2020-04-27	12.03	+103.2%

Table C.2. The ten worst trades, 2015–2026.

#	Stock	Bought	Buy price	Sold	Sell price	Return
1	PLUG	2021-02-16	65.01	2021-05-17	24.25	−62.8%
2	CDE	2019-12-30	8.01	2020-03-31	3.38	−58.0%
3	EGAN	2018-08-30	15.3	2018-11-29	7.46	−51.5%
4	VGZ	2016-07-14	1.6	2016-10-12	0.85	−47.1%
5	TTMI	2026-06-15	203.84	2026-09-15	115.83	−43.3%
6	PACK	2021-11-29	40.05	2022-03-01	23.66	−41.2%
7	TSSI	2025-01-17	12.19	2025-04-21	7.28	−40.6%
8	PAMT	2018-09-20	15.44	2018-12-20	9.25	−40.4%
9	LASR	2026-05-22	77.36	2026-08-24	46.67	−39.9%
10	PSIX	2025-09-04	87.4	2025-12-03	54.5	−37.9%

Table C.3. Every position held in the forward test, 19 September to 11 November 2026.

#	Stock	Bought	Buy price	Sold	Price	Return
1	DFTX	2026-08-06	44.39	2026-11-04	33.48	−24.9%
2	ATEX	2026-08-27	94.16	open	71.76	−23.9%
3	HSHP	2026-08-27	15.47	open	16.46	+6.3%
4	CLYM	2026-09-08	15.62	open	11.83	−24.3%
5	TWST	2026-09-15	134.04	open	137.84	+2.7%
6	HUM	2026-11-04	414.38	open	443.34	+6.9%

Table C.4. Every trade entered in the QISC out-of-sample window and closed by 11 November 2026.

#	Stock	Bought	Buy price	Sold	Sell price	Return
1	ETON	2025-01-13	13.29	2025-04-14	12.77	−4.4%
2	TSSI	2025-01-17	12.19	2025-04-21	7.28	−40.6%
3	NUTX	2025-02-26	57.0	2025-05-28	165.44	+188.9%
4	DX	2025-03-14	10.74	2025-06-13	9.83	−8.9%
5	AHR	2025-03-19	28.18	2025-06-18	34.24	+21.1%

#	Stock	Bought	Buy price	Sold	Sell price	Return
6	WGS	2025-04-14	99.94	2025-07-16	82.73	−17.5%
7	CME	2025-04-21	251.09	2025-07-22	263.47	+4.7%
8	FSCO	2025-05-28	5.87	2025-08-27	6.3	+6.7%
9	CAH	2025-06-13	158.8	2025-09-15	150.3	−5.5%
10	CNL	2025-06-26	10.47	2025-09-25	13.34	+26.7%
11	AHR	2025-07-16	36.4	2025-10-14	40.53	+10.9%
12	TDUP	2025-07-22	8.36	2025-10-20	9.26	+10.2%
13	PSIX	2025-09-04	87.4	2025-12-03	54.5	−37.9%
14	AHR	2025-10-23	42.29	2026-01-26	45.67	+7.7%
15	BE	2025-10-24	106.0	2026-01-27	141.0	+32.8%
16	AMLX	2025-10-24	14.52	2026-01-27	15.0	+2.7%
17	LEU	2025-10-27	391.15	2026-01-28	313.0	−20.1%
18	SNDK	2025-12-11	230.0	2026-03-16	687.52	+198.3%
19	LITE	2026-02-02	405.49	2026-05-04	986.88	+142.9%
20	AXTI	2026-02-03	21.8	2026-05-05	106.78	+388.2%
21	HYMC	2026-02-04	43.46	2026-05-06	37.58	−13.8%
22	LASR	2026-02-09	51.2	2026-05-11	75.46	+46.9%
23	WDC	2026-03-16	282.63	2026-06-15	618.58	+118.4%
24	ORKA	2026-05-06	69.62	2026-08-06	96.74	+38.2%
25	LITE	2026-05-22	965.15	2026-08-24	823.6	−14.8%
26	LASR	2026-05-22	77.36	2026-08-24	46.67	−39.9%
27	SYRE	2026-06-08	75.53	2026-09-08	85.0	+12.0%
28	TTMI	2026-06-15	203.84	2026-09-15	115.83	−43.3%
29	DFTX	2026-08-06	44.39	2026-11-04	33.48	−24.9%